VECTOR ADDITION

2.1 Program Outcomes (POs) Addressed by the Activity

At the end of the term, the students must be able to

a. Ability to apply knowledge of computing fundamentals, computing specialization, and mathematics, science, and domain knowledge appropriate for the computing models.
b. Ability to function effectively as a leader and as a member or leader in diverse teams and in multidisciplinary settings.
c. Ability to communicate effectively with the computing community and with society at large about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions.

2.2 Activity's Intended Learning Outcomes (AILOs)

At the end of this activity, the student should be able to

a. represent vector quantities graphically,
b. add vectors using the graphical method,
c. determine the equilibrant force and the resultant force.

2.3 Objectives of the Activity

At the end of this activity, the student should be able to

a. to determine the magnitude and direction of the equilibrant and resultant vectors using the force table, and
b. to determine the magnitude and direction of the resultant vectors using the polygon method (graphical method).

2.4 Principle of the Activity

A vector quantity is represented by magnitude and direction. Force, displacement, and acceleration are examples of vectors. When performing operations involving vectors, the direction should also be considered. The sum of two or more vectors is called the **resultant** vector. We can add vectors using the polygon (graphical) or component (algebraic) method.

In this experiment, we will determine the Resultant vector using the force table and the polygon method. **The resultant is the negative of the equilibrant, that is, they have equal magnitudes but opposite directions.**

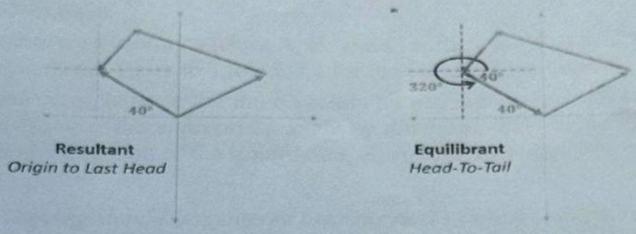

Figure 2.1. Sample resultant and equilibrant graphical representation.

2.5 Materials/Equipment

Qty	Item	Group	Class
1 (set)	force table equipment	Measuring tool	0
1 (set)	weights	Others	0
1 m	string	Others	0
1 pc	protractor	Measuring tool	0
1 pc	ruler	Measuring tool	0

2.6 Procedure/s

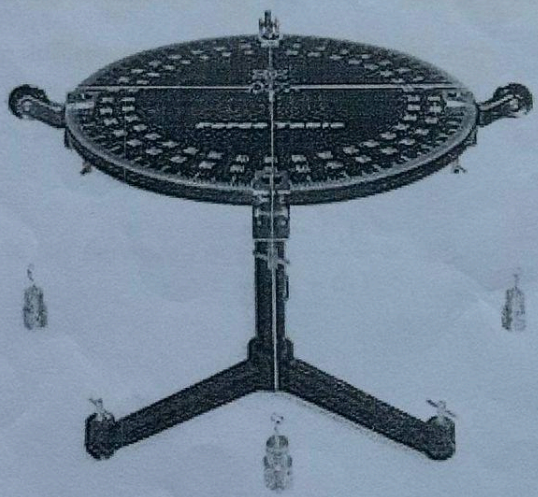

Figure 2.2. Experimental setup for force table with hanging masses attached to the strings.

The force table has four pulleys for the four weights to be attached. Three will be used for the three forces then the other one for the equilibrant.

2.6.1 Getting the Resultant Using the Force Table

a. Set up the apparatus as shown in Figure 2.2. Tie four strings of equal length to the ring provided with the force table.

b. Hang weights representing vectors A, B, and C at the end of each of the strings acting in the assigned directions. Refer to Table 2.7.1 for the values of the vectors.

c. By trial and error, balance the three vectors by placing counterweights on the fourth unused string and locating the appropriate angle or direction. This is the **Equilibrant.** Record the magnitude and direction of the Equilibrant, **E**, and the Resultant, $\mathbf{R_1}$, in Table 2.7.1.

***Remark**: The ring should be centered over the port when the system is in equilibrium. Pull the string slightly to one side and let it go. The ring should return to the center. If not, adjust the fourth vector's magnitude and direction until the ring returns to the center after pulling it to one side.*

2.6.2 Getting the Resultant Using the Polygon Method

a. Use the same vectors A, B, and C set in Part I.

b. Use the tip-to-tail method of graphing. Use a scale of 0.49 N to 1 inch.

c. Using a ruler and a protractor, sketch vector A. The tail of vector A should be at the origin. At the tip of vector A (this serves as the origin of the next vector), connect vector B. Then connect vector C at the tip of vector B.

d. Draw a line connecting the origin and the tip of vector C and an arrow tip pointing to vector C. This is the Resultant vector. Measure the length of the line representing the resultant using a ruler and convert it to the unit it represents. Record it in Table 2.7.2 as the magnitude of the R_1.

e. Measure the direction of the Resultant using a protractor. This is denoted as θ_2. Record it in Table 2.7.2.

f. Compute the percentage difference using the following equations:

$$\%\,Difference_R = \frac{|R_2 - R_1|}{\frac{1}{2}(R_2 + R_1)} \times 100$$

$$\%\,Difference_\theta = \frac{|\theta_2 - \theta_1|}{\frac{1}{2}(\theta_2 + \theta_1)} \times 100$$